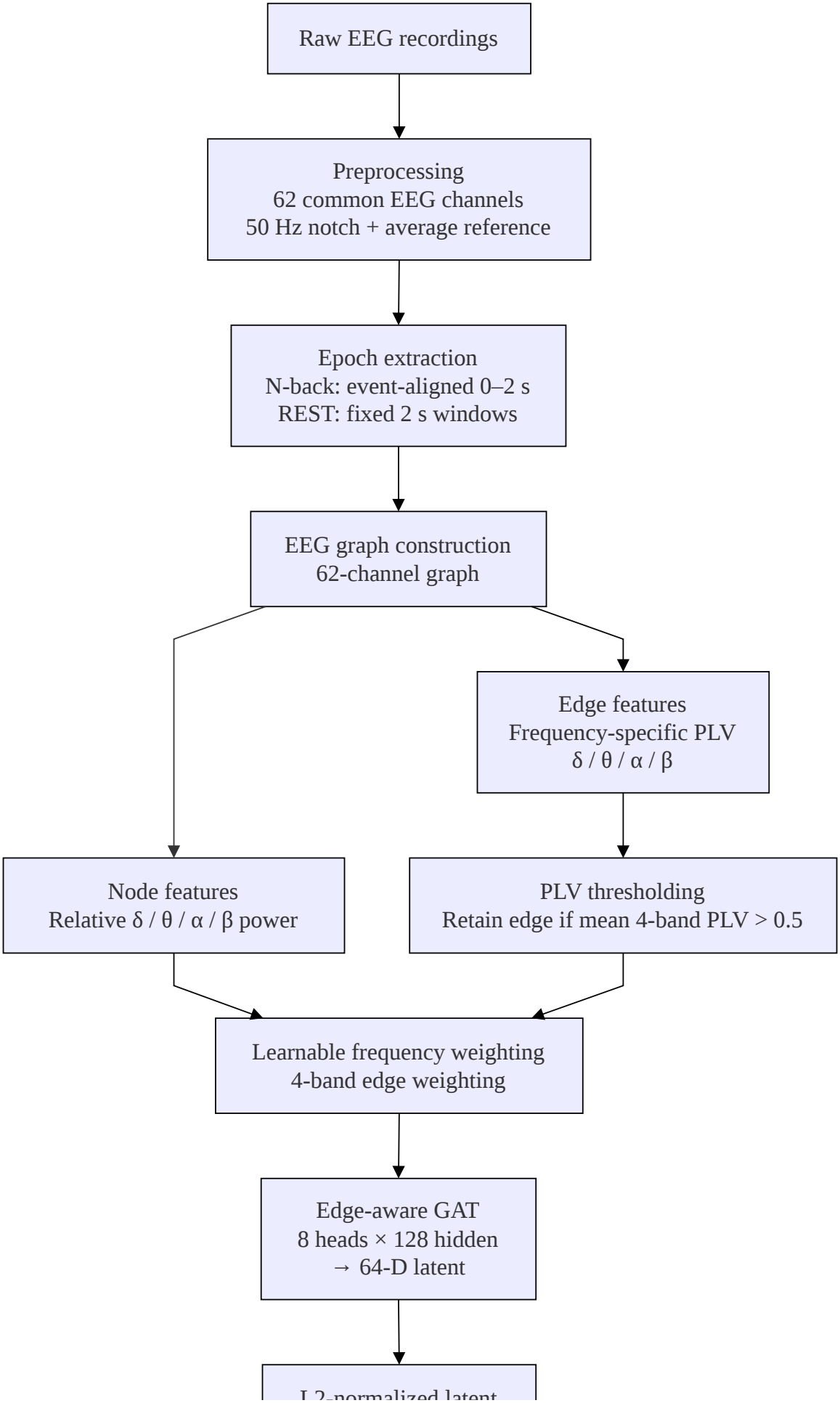
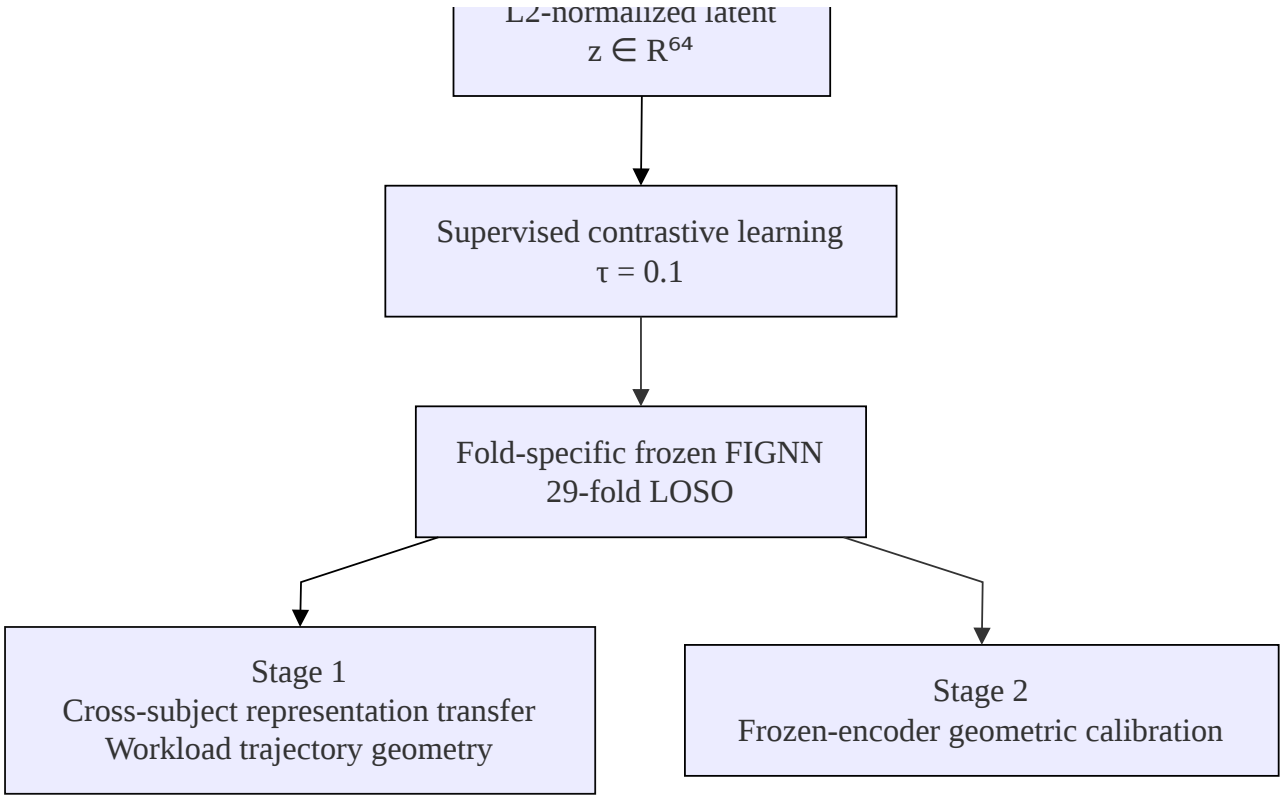

Raw EEG recordings
Preprocessing
62 common EEG channels
50 Hz notch + average reference
Epoch extraction
N-back: event-aligned 0–2 s
REST: fixed 2 s windows
EEG graph construction
62-channel graph
Edge features
Frequency-specific PLV
δ / θ / α / β
Node features
Relative δ / θ / α / β power
PLV thresholding
Retain edge if mean 4-band PLV > 0.5
Learnable frequency weighting
4-band edge weighting
Edge-aware GAT
8 heads × 128 hidden
→ 64-D latent
L2-normalized latent


L2-normalized latent
z ∈ R⁶⁴
Supervised contrastive learning
τ = 0.1
Fold-specific frozen FIGNN
29-fold LOSO
Stage 1
Cross-subject representation transfer
Workload trajectory geometry
Stage 2
Frozen-encoder geometric calibration